UNIVERSITÀ DEGLI STUDI FIRENZE

Scuola di Ingegneria

HR EXCELLENCE IN RESEARCH

Development of a semi-automatic annotation system of anomalies in thermal videos

Computer Engineering Bachelor’s degree

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Supervisor: Marco Bertini

The Context

ANOMALY DETECTION

- Computer Vision.
- Definition of anomalies with regards to normalcy.
- Anomaly Detection is dependent on context and environment.

THERMAL VIDEOS

- Technology able to measure the intensity of infrared radiations.
- Anomalies defined as "hot-spot": $\Delta T(ROI, REF)$

ANNOTATION

Pre-training phase for computer vision models.

1. Frame annotation – identify and label.
2. Tracking and propagation – keeping track of the object in sequences of consecutive frames.

Costs and Challenges

ANNOTATION

Model training needs:

- Vaste and diverse datasets
- Specialisation for specific or rare contexts

Annotation is a costly process in terms of:

- time
- resources

THERMAL VIDEOS

Compared to optic imagery (RGB), the difficulties tied to thermal videos are linked to:

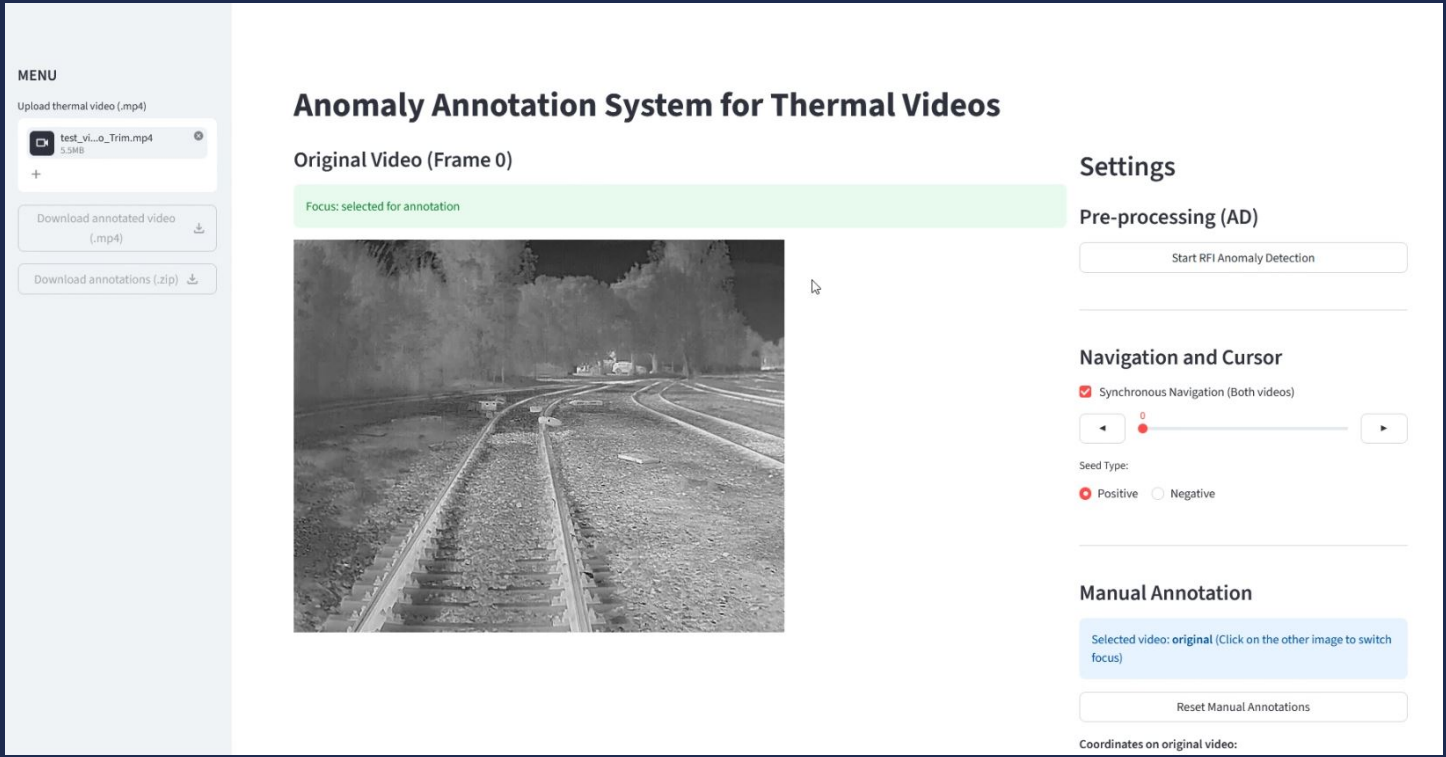
- Lower resolution
- Lower signal/noise ratio
- Blurred details
- Low contrast
- Lack of texture
- Lack of training

SAFETY-CRITICAL DOMAIN

It is not possible to fully automate the annotation process, especially in the Anomaly Detection field.

In safety-critical domains, as railways infrastructures, supervision is necessary.

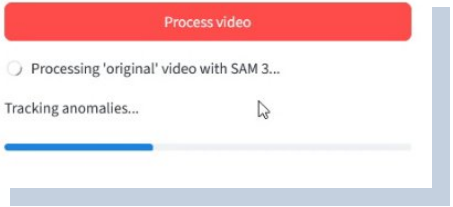
THE SYSTEM


MENU
Upload thermal video (.mp4)
test_vi...o_Trim.mp4
5.5MB
Download annotated video (.mp4)
Download annotations (.zip)
Anomaly Annotation System for Thermal Videos
Original Video (Frame 0)
Focus: selected for annotation
Settings
Pre-processing (AD)
Start RFI Anomaly Detection
Navigation and Cursor
Synchronous Navigation (Both videos)
0
Seed Type:
Positive
Negative
Manual Annotation
Selected video: original (Click on the other image to switch focus)
Reset Manual Annotations
Coordinates on original video:

Addressing the Challenges

Focusing manual labot in prcise tasks of revision and inspection.

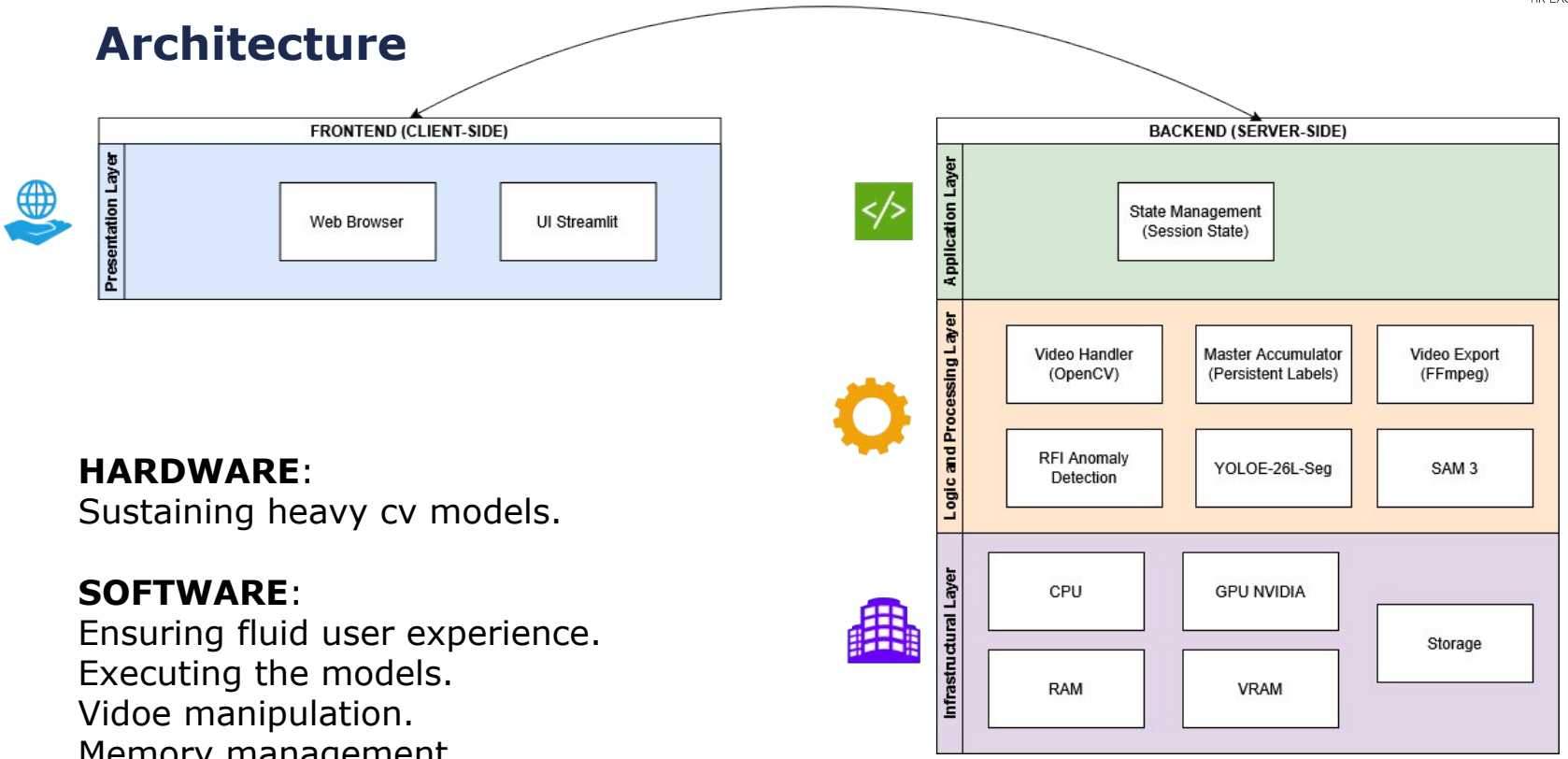
More demanding jobs are assigned to the computer vision models.



Human-In-The-Loop Paradigm:

- Active Learning: the models takes control.
- Interactive Machine Learning: control is shared between model and human.
- Machine Teaching: control in the hands of the human.

Architecture



HARDWARE:
Sustaining heavy cv models.

SOFTWARE:
Ensuring fluid user experience.
Executing the models.
Vidoe manipulation.
Memory management.

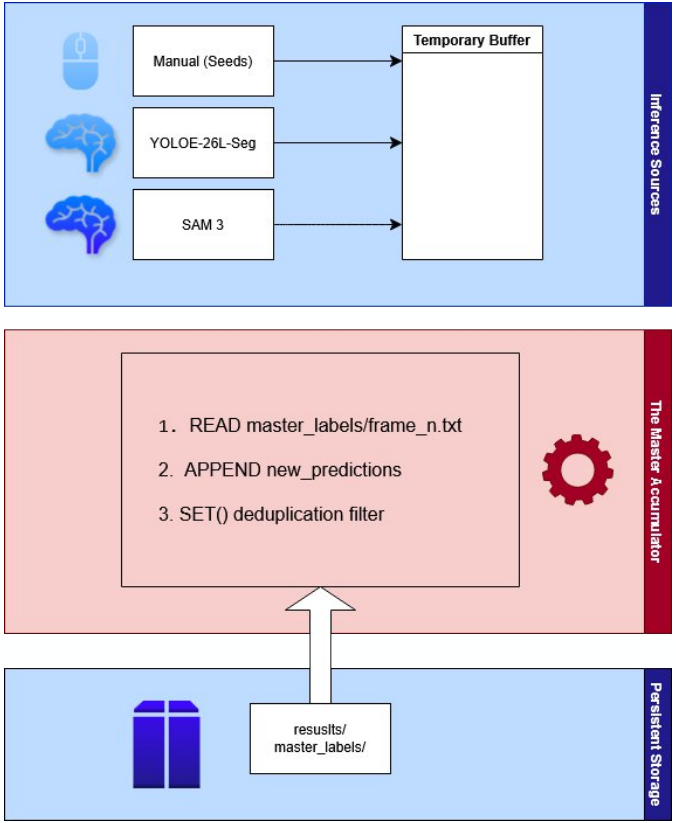
Memory Management

Persistent Label Accumulation

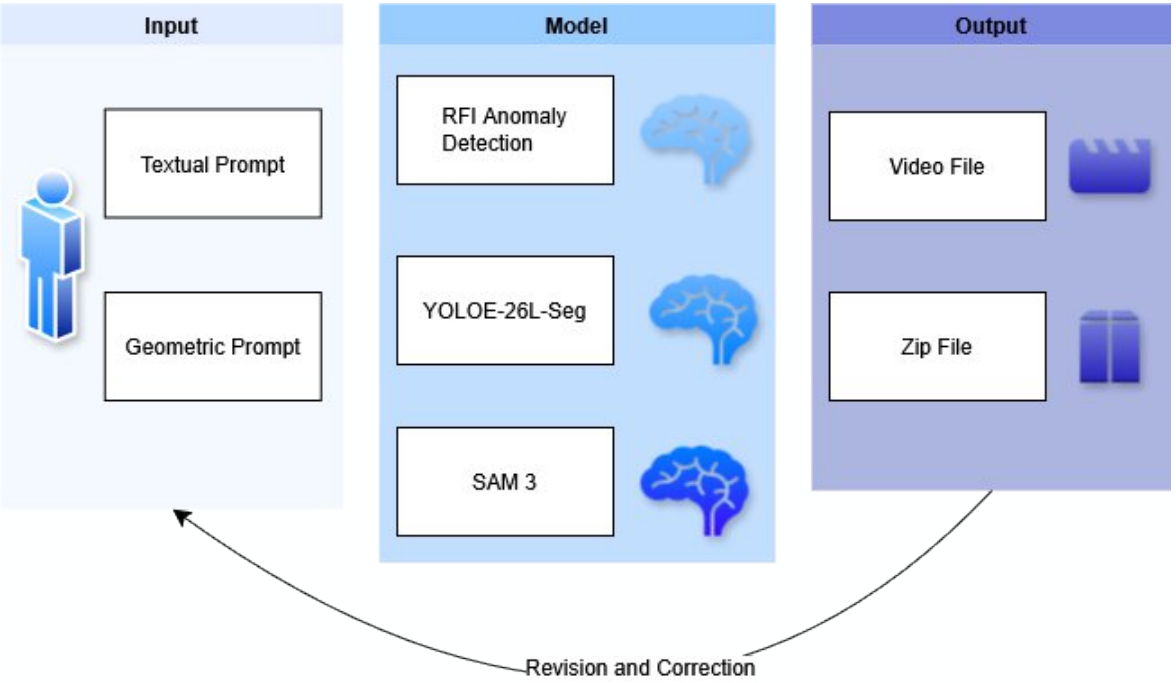
YOLO-Seg-Polygon Format

[class, coordinates]

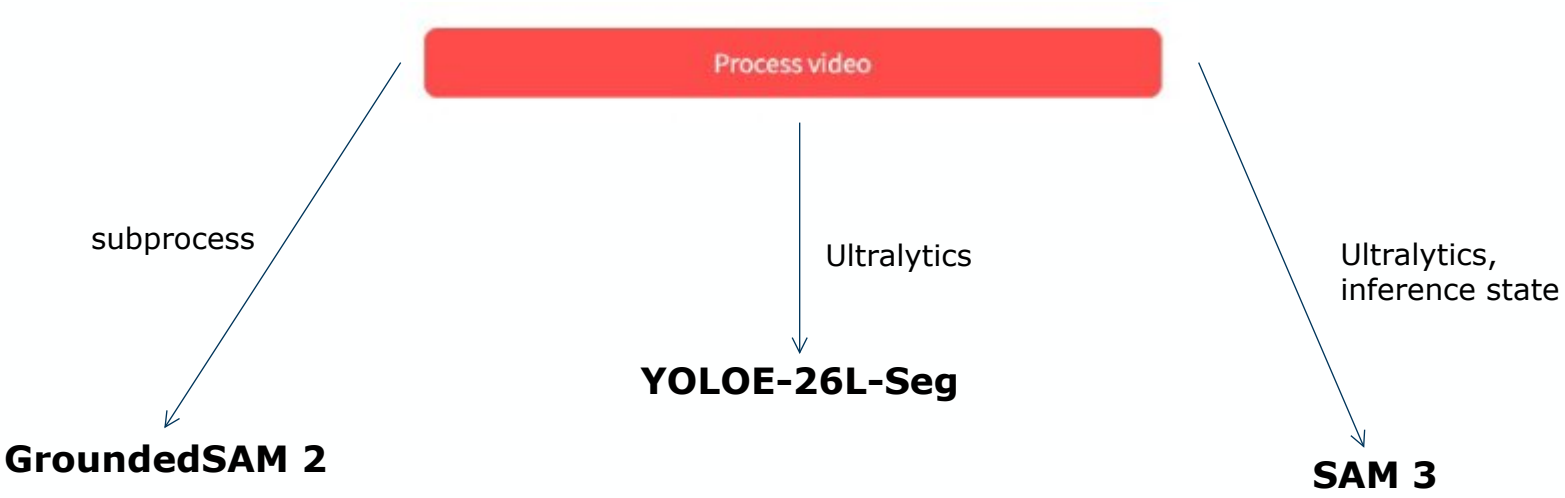
Expansion .txt



User Workflow



Models Intergration



RFI Anomaly Detection

- GroundingDino and SAM 2.
- Railways infrastructures
- Pre-screening to find “suspect frames”

Integration via the subprocess library

YOLOE-26L-Seg

- YOLO family: You Only Look Once
- “E” – extends open-vocabulary textual input
- “26” – structural changes (end-to-end)
- “L” – Large-Scale configuration of the neural network
- “Seg” - segmentation

Integration through Ultralytics

SAM 3

- State-of-the-Art for segmentation models
- Promptable Visual Segmentation
- Promptable Concept Segmentation
- Dual structure
- At the base of the application workflow
- Integration through:
  - Ultralytics for the PVS approach
  - Inference state for the PCS approach

Results and Considerations

SYSTEM

- Laying the basis to address the difficulties derived from the thermal domain and annotation in general
- Supporting heavy models
- Information persistence in favour of the UX and from a functional point of view to update annotations during the process
- UI follows the application's *workflow* top-down, making its use intuitive.

MODELS

The main weakness for both models is caused by the thermal domain.

YOLOE-26L-Seg	SAM 3
It finda most challenging the act of associating open-vocabulary concepts to objects in the thermal domain.	The implementation of mask generation through geometric inputs aids the segmentation process when PCS fails.
It shows its usefulness in the initial processing phase to identify regions of interest (ROI).	Even though the lack of defined outlines challenges its ability toi create precise masks.

Results and Considerations – IoU

1. Choosing reference image
2. Manual annotation
3. Segmentations done by the models
4. Calculating Intersection over Union between Ground Truth and models predictions

	YOLOE-26L-Seg	SAM 3 (PCS)	SAM 3 (PVS)
IoU	30.54%	71.40%	43.22%

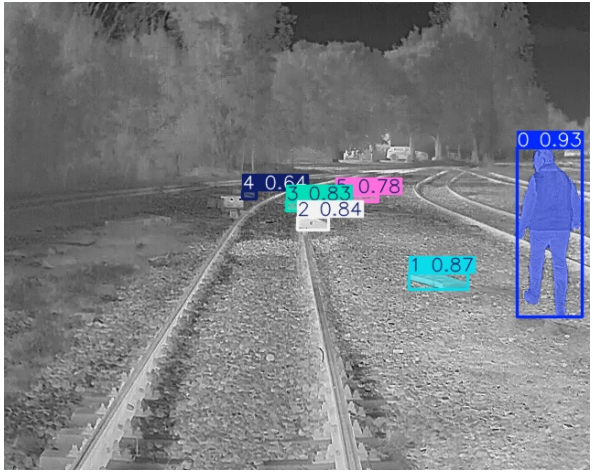
Results and Considerations – YOLOE-26L-Seg

Ground Truth - annotazione manuale

YOLOE-26L-Seg segmentation

Results and Considerations – YOLOE-26L-Seg

Ground Truth - annotazione manuale



SAM 3 PCS segmentation

Results and Considerations – YOLOE-26L-Seg

Ground Truth - manual annotationn

PVS asegmentation

Expansions and Improvements

- Introduction of “brush” and “eraser” tools for final refining.
- Video navigation through the first frames masks are generated.
- Bringing SAM 3 ability to maintain unique identifiers of the objects associated to the masks to user level, connecting them to the seeds.
- Emphising the distinction between first pass of the segmentation models and the refining anc correction cycles.
- Studying the possibility to add a database to better organise the process of dataset creation.

Thank you for your attentionfo

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